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BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

It is an organised, structured data stored is a computer system.

Database are essential because they help store large amout of data efficiently and retrieve it quickly

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

warehouse store large used for analysis and decision making

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- A data lake is a storage system that holds raw data in its original formats, whether structured or not

① Datawarehouse - structured & organised
② Data lake - raw, unprocessed data.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

- A data mart is a smaller section of data warehouse that focuses on a specific department or business area.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured - organised in table
Semi-Structured - JSON
unstructured - Images, Video

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- Data modelling is the process of designing how data is organised and structured,
- improve data quality
- Reduce redundancy

Types - Conceptual, logical and Physical

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- is the Process of analysing large datasets to find patterns and trends
- Retail Stores analysing buying patterns

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

DBMS - is Software used to create and manage databases

MySQL - Websites
PostgreSQL - advanced analytics
Oracle - large enterprises
MongoDB - flexible NOSQL data

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- Extract - Collect data
- Transform - Clean and Format
- Load - Store in database

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

CSV- Simple files
~~Joson~~

~~JSO~~
JSON - APIS
Parquest- big data
Avro - fast processing

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

AI - Is the ability of machines to perform task normally require human intelligence

- Narrow AI - ChatGPT
General AI - human-like intelligence

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

ML - allows computers to learn from data instead of being explicity programmed

* Supervised - uses labelled data
* Unsupervised - Find patterns
* Reinforment - learns through rewards

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

Deep learning is a type of ML that uses neural Networks

Neural networks
→ They mimic the human brain & process information in layers

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP - allows ~~Com~~ Computer to understand human language

Chatbot, google Translate & Voice assistant's

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- LLM is an AI trained on large text dataset

ChatGpt
Claude

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

Generative AI Create content such as text images and ~~vid~~ video
- ChatGpT uses text
- DALLE use images
- Midjourney - artwork

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

A training dataset is the data used to train an AI model

Impurtance - good data leads to accurate model

Risk - Biased leads to unfair results

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer vision allows machines to interpret images
face recognition
Self-driving cars

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

API allows different systems to communicate.

Analogy - like a waiter taking your order to the kitchen

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

An API key is a secret code used to access an API

If exposed the risk could lead to security issues

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

Rest stand for Representational State Trasfer

Get - retrieve data
Post - Creat data
PUT - update data
Delete - remove data.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON - is a format used to store and exchange data

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- it is the skill of writting effective instructions of AI

BAD - Explain AI
GOOD - Explain AI is simple terms with exemples

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

The map function applies a function to each ~~elem~~ elements in a list

Map (x - x * 2

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Using internet based servers

Iaas - AWS
Paas -> Coogle App engine
Saas -> Gmail

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Track changes in code

Git: 1 Commit - Save changes
2 Branch - Create version.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

A tool for writing and running code

Python

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Pytho is a programming language for data analysis

libraries
- Pandas
- Numpy
- Matplotlib
- Scikit-learn

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Protecting personal data
GDPR
- A law in Europe that protect user data

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Using data responsibly
Explain
* Selling personal data
* AI discrimination
* fake data

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

Managing data properly
Include
* Policies, security & Data quality

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

Bias occurs when AI produces unfair result
E.g Hiring systems favouring certain groups

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Role	Description
Data Analysis	Analyses data
Data Scientist	Builds model
Data Engineer	Builds data systems
AI Engineer	Develops AI system

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

BI – help organisations make decisions.

Tool
* Power BI
* Tableau

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

I found ML interesting because it allows computer to learn from data and improve over time.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

I found data Modelling difficult, but I used online tutorails and video to understand it better

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data & AI will improve decision making and Create job opportunities
Businesses can Predict Customer behaviour

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. IBM (2023) What Data Science.
2. AWS (2023). What is a Database
3. Microsoft (2023) AI Overview.
4.
5.
6.
7.
8.

Good luck! We believe in you.
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